



2 Real Analysis 2023

Tutorial 1

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Rational and irrational numbers

Problem 1 (a) Prove the following statement: If x is rational and y is irrational, then $x + y$ is irrational.

(b) Why is the statement “If x is rational and y is irrational, then xy is irrational” false? Modify the statement to make it true. Prove your modified statement.

The principle of mathematical induction

Problem 2 Prove the following statements using the principle of mathematical induction.

(a) For $x \in \mathbb{R}$, $x \neq 1$ and $n = 0, 1, 2, \dots$ we have

$$(1+x)(1+x^2)(1+x^4) \dots (1+x^{2^n}) = \frac{1-x^{2^{n+1}}}{1-x}.$$

(b) Let $(a_n)_{n=1}^\infty$ be the sequence defined recursively by

$$\begin{aligned} a_1 &= 1, & a_2 &= 5, \\ a_{n+2} &= 5a_{n+1} - 6a_n, & n &\in \mathbb{N}. \end{aligned}$$

Prove that $a_n = 3^n - 2^n$, $n \in \mathbb{N}$.

Irrationality of $\sqrt{2}$

Problem 3 (a) We denote by $\sqrt{2}$ the non-negative number r with $r^2 = 2$. We have proven in the lecture that $\sqrt{2} \notin \mathbb{Q}$. Below is a sketch of the proof:

Suppose that $\sqrt{2} \in \mathbb{Q}$. Consider the set $S = \{q \in \mathbb{N} : \sqrt{2}q \in \mathbb{Z}\}$. Then S is a nonempty subset of $\mathbb{N}$, and therefore S has a smallest element, say q_0 . Put $q_1 = (\sqrt{2} - 1)q_0$. Then $q_1 \in \mathbb{Z}$ and $0 < q_1 < q_0$. Moreover, $\sqrt{2}q_1 \in \mathbb{Z}$, and this is a contradiction. The contradiction shows that $\sqrt{2}$ is irrational.

Answer the following questions:

(i) Why is S nonempty?

- (ii) Why is there a smallest element q_0 ?
 - (iii) Why is $q_1 \in \mathbb{Z}$?
 - (iv) Why is $0 < q_1 < q_0$?
 - (v) Why is $\sqrt{2}q_1 \in \mathbb{Z}$?
 - (vi) Explain how we obtain a contradiction.
- (b) Modify the above proof to show that $\sqrt{5}$ is irrational.
- (c) Clearly, $\sqrt{4} = 2$ is rational. If we try to modify the above proof to consider $\sqrt{4}$, it will not work. Why?